

Classical Mechanics 1

Quiz 1

1. [Taylor 1.11 ★] The position of a moving particle is given as a function of time t to be

$$\mathbf{r}(t) = \hat{\mathbf{x}} b \cos(\omega t) + \hat{\mathbf{y}} c \sin(\omega t),$$

where b , c and ω are constants. Describe the particle's orbit.

2. [Taylor 1.39 ★★] A ball is thrown with initial speed v_0 up an inclined plane. The plane is inclined at an angle ϕ above the horizontal, and the ball's initial velocity is at an angle θ above the plane. Choose axes with x measured up the slope, y normal to the slope, and z across it. Write down Newton's second law using these axes and find the ball's position as a function of time. Show that the ball lands a distance

$$R = \frac{2v_0^2 \sin \theta \cos(\theta + \phi)}{g \cos^2 \phi}$$

from its launch point. Show that for given v_0 and ϕ , the maximum possible range up the inclined plane is

$$R_{\max} = \frac{v_0^2}{g(1 + \sin \phi)}.$$